Signals and Systems Lecture-10: Fourier Transform (1)

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Introduction

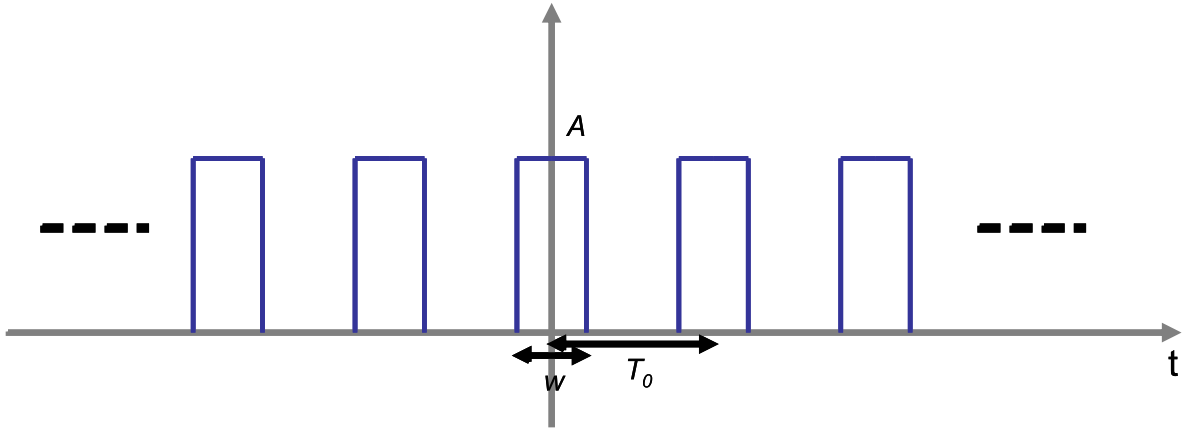
- Consider an aperiodic continuous-time (CT) signal $x(t)$.
  - This signal can be viewed as a periodic signal with period $T_0 \rightarrow +\infty$.
- When plotting the spectrum of periodic signals, the harmonic components are $f_0 = \frac{1}{T_0}$ Hz apart.
- As $T_0 \rightarrow +\infty$, $f_0 \rightarrow 0$ and the harmonic components are spaced closer and closer in frequency.
- Consequently:

$$\text{Periodic Signals} \Rightarrow \text{Discrete Spectrum}$$
$$\text{Aperiodic Signals} \Rightarrow \text{Continuous Spectrum}$$

- The Fourier Transform is used to study the spectrum of aperiodic signals (as well as periodic signals).

Motivating Example (1)

- Consider the periodic signal obtained from repeating the signal $A\text{rect}\left(\frac{t}{w}\right)$ with a period T_0.



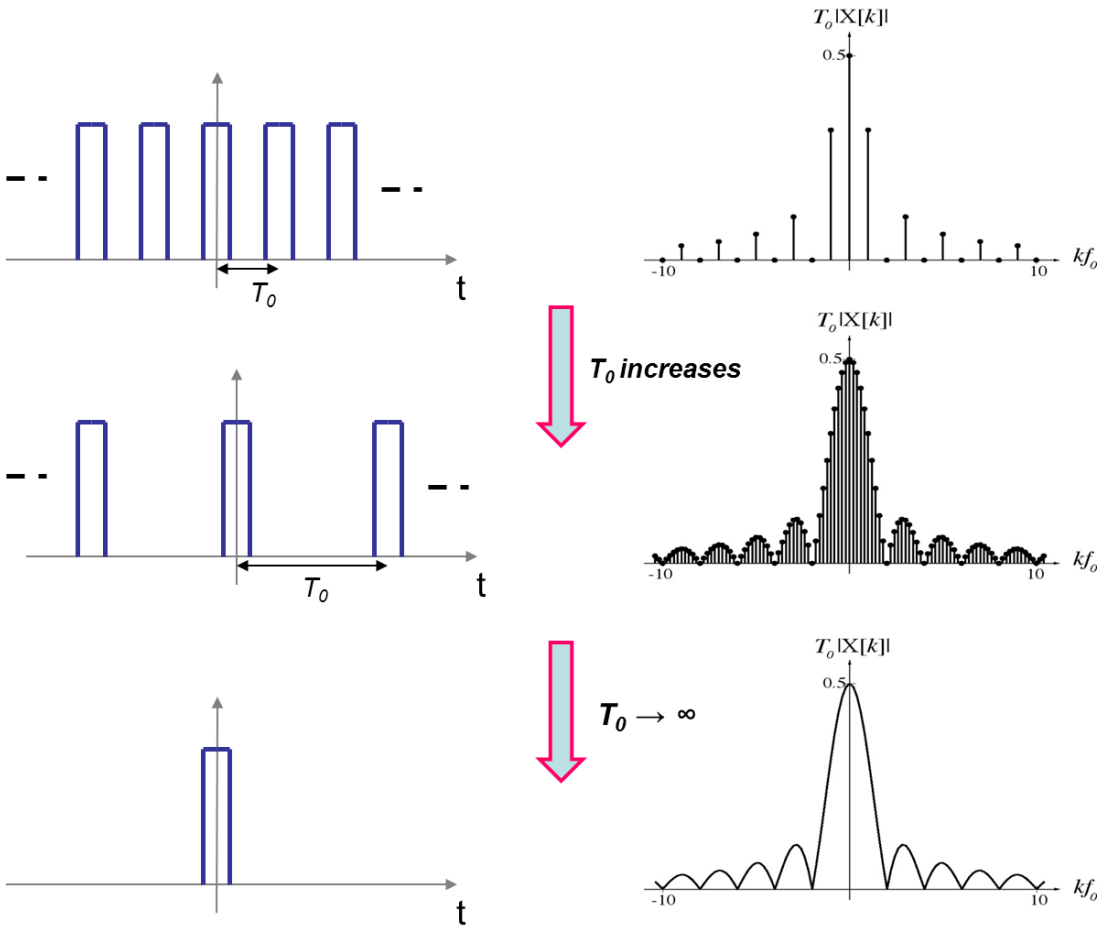
- We have seen in previous chapters that the Fourier series coefficients of this signal are given by:

$$X[k] = A\frac{w}{T_0}\text{sinc}\left(k\frac{w}{T_0}\right) \quad \Rightarrow \quad T_0X[k] = Aw\text{sinc}\left(k\frac{w}{T_0}\right)$$

- Let us see what will happen when increasing T_0 while w is held constant.

Motivating Example (2)

- Plotting $T_0 X[k]$ as a function of $kf_0 = \frac{k}{T_0}$:



- At the limit, as the period approaches infinity, $T_0 X[k]$ will approach a continuous function of the frequency f.

Fourier Transform: Definition

Fourier Transform of CT signals:

- The Fourier Transform (FT) is defined by:

$$\mathcal{F}: \; x(t) \rightarrow X(f)$$

$$X(f) = \mathcal{F}(x(t)) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi ft} \mathrm{d}t$$

- The inverse Fourier Transform is defined by:

$$\mathcal{F}^{-1}: \; X(f) \rightarrow x(t)$$

$$x(t) = \mathcal{F}^{-1}(X(f)) = \int_{-\infty}^{+\infty} X(f) e^{j2\pi ft} \mathrm{d}f$$

Total area integrals using FT

Total Area Theorems:

- Total area under $x(t)$:
  - Since:

$$X(f) = \int_{-\infty}^{+\infty} x(t)e^{-j2\pi ft}\mathrm{d}t$$

then for $f = 0$:

$$X(0) = \int_{-\infty}^{+\infty} x(t)\mathrm{d}t$$

- Total area under $X(f)$:
  - Since:

$$x(t) = \int_{-\infty}^{+\infty} X(f)e^{j2\pi ft}\mathrm{d}f$$

then for $t = 0$:

$$x(0) = \int_{-\infty}^{+\infty} X(f)\mathrm{d}f$$

Fourier Transform of Real Signals (1)

- **Proposition**: For real signals, the Fourier transform satisfies the following relation:

$$X(-f) = X^*(f)$$

- *Proof:*
  - Since $X(f) = \int_{-\infty}^{+\infty} x(t)e^{-j2\pi ft}\mathrm{d}t$:

$$X(-f) = \int_{-\infty}^{+\infty} x(t)e^{j2\pi ft}\mathrm{d}t$$

  - On the other hand:

$$X^*(f) = \left[\int_{-\infty}^{+\infty} x(t)e^{-j2\pi ft}\mathrm{d}t\right]^* = \int_{-\infty}^{+\infty} x^*(t)\left[e^{-j2\pi ft}\right]^*\mathrm{d}t$$
$$= \int_{-\infty}^{+\infty} x(t)e^{j2\pi ft}\mathrm{d}t$$

    - where $x^*(t) = x(t)$ since $x(t)$ is real.
  - Consequently, $X(-f) = X^*(f)$.

Fourier Transform of Real Signals (2)

- Therefore, for real signals:

$$X(-f) = X^*(f)$$

- For the magnitude spectrum $|X(f)|$:

$$|X(-f)| = |X^*(f)| = |X(f)|$$

implying that $|X(f)|$ is even.

- For the phase spectrum $\arg[X(f)]$:

$$\arg[X(-f)] = \arg[X^*(f)] = -\arg[X(f)]$$

implying that $\arg[X(f)]$ is odd.

Fourier Transform of Real Signals (3)

- For real signals, the FT can be written as:

$$\begin{aligned} X(f) &= \int_{-\infty}^{+\infty} x(t)e^{-j2\pi ft}\mathrm{d}t \\ &= \int_{-\infty}^{+\infty} x(t)\cos(2\pi ft)\mathrm{d}t - j\int_{-\infty}^{+\infty} x(t)\sin(2\pi ft)\mathrm{d}t \\ &\triangleq \Re[X(f)] + j\Im[X(f)] \end{aligned}$$

  - For the real part of $X(f)$:

$$\Re[X(-f)] = \int_{-\infty}^{+\infty} x(t)\cos(-2\pi ft)\mathrm{d}t = \int_{-\infty}^{+\infty} x(t)\cos(2\pi ft)\mathrm{d}t = \Re[X(f)]$$

    implying that $\Re[X(f)]$ is an even function of f.
  - For the imaginary part of $X(f)$:

$$\Im[X(-f)] = -\int_{-\infty}^{+\infty} x(t)\sin(-2\pi ft)\mathrm{d}t = \int_{-\infty}^{+\infty} x(t)\sin(2\pi ft)\mathrm{d}t = -\Im[X(f)]$$

    implying that $\Im[X(f)]$ is an odd function of f.

Fourier Transform of Real Signals (4)

- If $x(t)$ is even, then $x(t)\sin(2\pi ft)$ is odd and:

$$\Im[X(f)] = -\int_{-\infty}^{+\infty} x(t)\sin(2\pi ft)\mathrm{d}t = 0$$

  Therefore, the Fourier Transform of a real and even signal is real and even.

- If $x(t)$ is odd, then $x(t)\cos(2\pi ft)$ is odd and:

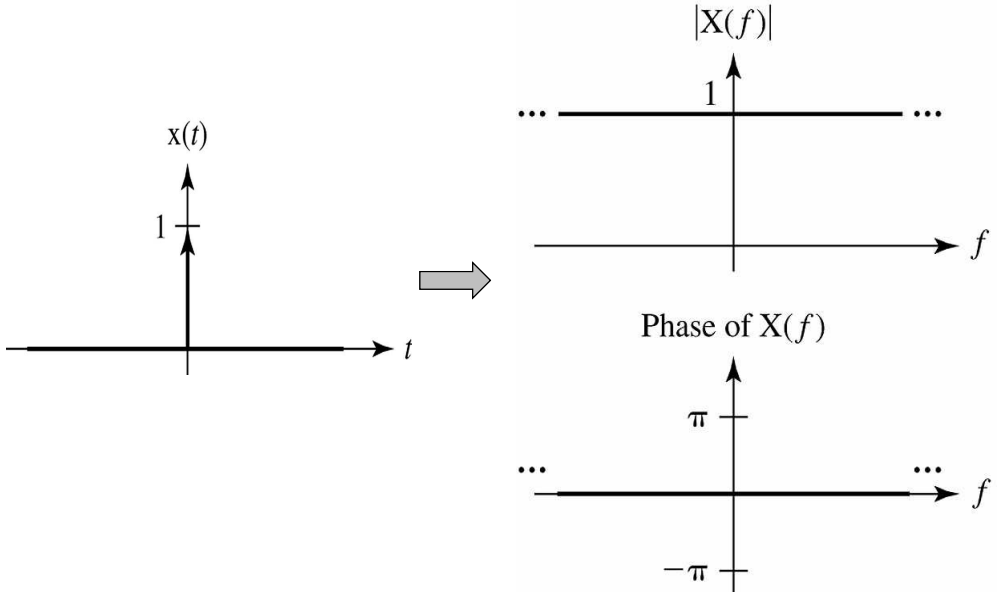
$$\Re[X(f)] = \int_{-\infty}^{+\infty} x(t)\cos(2\pi ft)\mathrm{d}t = 0$$

  Therefore, the Fourier Transform of a real and odd signal is imaginary and odd.

Fourier Transform of the unit impulse (1)

- The Fourier Transform of the unit impulse $x(t) = \delta(t)$ is:

$$X(f) = \int_{-\infty}^{+\infty} \delta(t) e^{-j2\pi ft} \mathrm{d}t = \int_{-\infty}^{+\infty} \delta(t) \left[e^{-j2\pi ft} \right]_{t=0} \mathrm{d}t = \int_{-\infty}^{+\infty} \delta(t) \mathrm{d}t = 1$$



- Consequently (applying the inverse Fourier Transform):

$$\delta(t) = \int_{-\infty}^{+\infty} e^{j2\pi ft} \mathrm{d}f$$

Fourier Transform of the unit impulse (2)

- Check the total area integral properties.
  - On one hand:

$$\int_{-\infty}^{+\infty} x(t)\mathrm{d}t = \int_{-\infty}^{+\infty} \delta(t)\mathrm{d}t = 1$$

  - Since $X(f) = 1$ for all values of f, then:

$$X(0) = 1$$

  - Consequently: $\int_{-\infty}^{+\infty} x(t)\mathrm{d}t = X(0) = 1$.
- In the same way:
  - On one hand:

$$\int_{-\infty}^{+\infty} X(f)\mathrm{d}f = \int_{-\infty}^{+\infty} 1\mathrm{d}f = +\infty$$

  - Since $x(t) = \delta(t)$, then for $t = 0$:

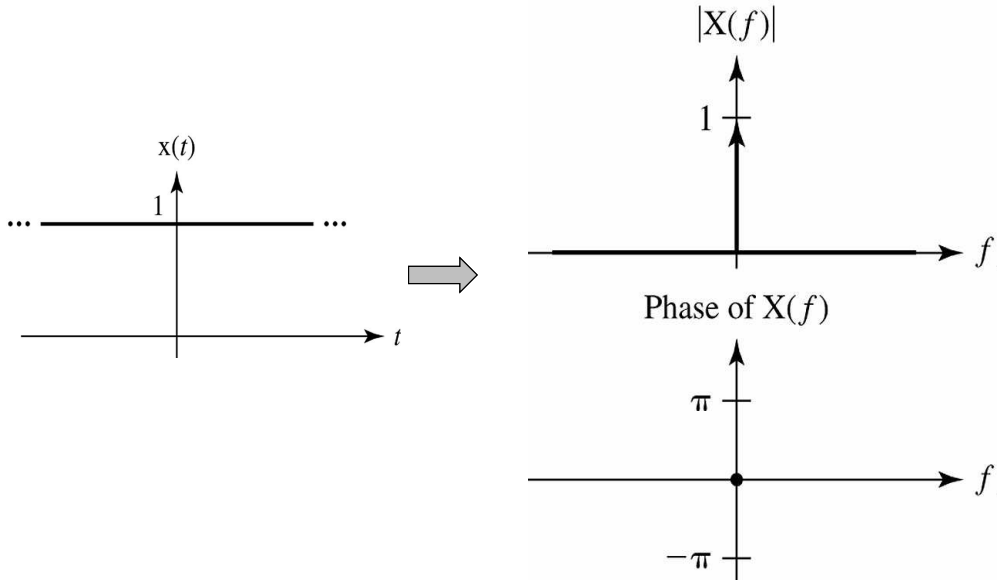
$$x(0) = +\infty$$

  - Consequently: $\int_{-\infty}^{+\infty} X(f)\mathrm{d}f = x(0) = +\infty$.

Inverse Fourier Transform of the unit impulse

- The Inverse Fourier Transform of $X(f) = \delta(f)$ is:

$$x(t) = \int_{-\infty}^{+\infty} \delta(f) e^{j2\pi ft} \mathrm{d}f = \int_{-\infty}^{+\infty} \delta(f) \left[e^{j2\pi ft} \right]_{f=0} \mathrm{d}f = \int_{-\infty}^{+\infty} \delta(f) \mathrm{d}f = 1$$



- In the same way: for $X(f) = \delta(f - f_0)$:

$$x(t) = \int_{-\infty}^{+\infty} \delta(f - f_0) e^{j2\pi ft} \mathrm{d}f = e^{j2\pi f_0 t}$$

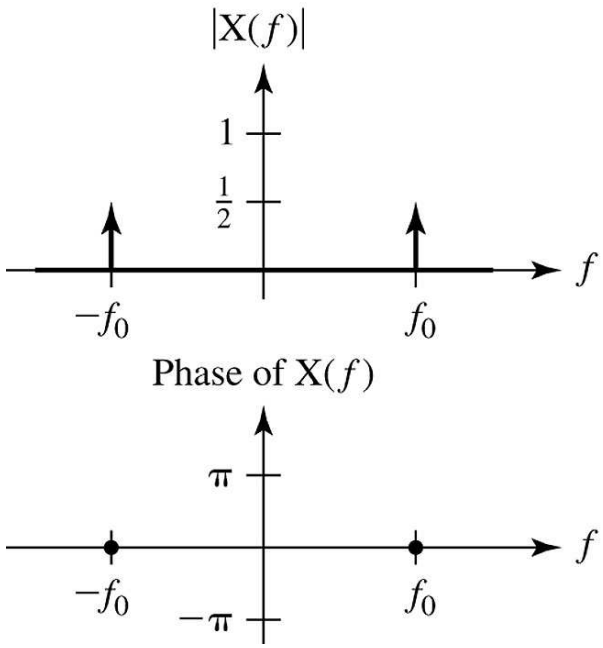
Fourier Transform of the cosine function

- The Fourier transform of $\cos(2\pi f_0 t)$ is:

$$x(t) = \cos(2\pi f_0 t) = \frac{1}{2}e^{j2\pi f_0 t} + \frac{1}{2}e^{-j2\pi f_0 t}$$

$$\Rightarrow \quad X(f) = \frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$$

- $x(t)$ is real and even and $X(f)$ is real and even.
- $|X(f)|$ is even and $\arg[X(f)]$ is odd.
- $x(0) = \cos(0) = 1$ and $\int_{-\infty}^{+\infty} X(f)\mathrm{d}f = \frac{1}{2} + \frac{1}{2} = 1$ implying that $x(0) = \int_{-\infty}^{+\infty} X(f)\mathrm{d}f$.



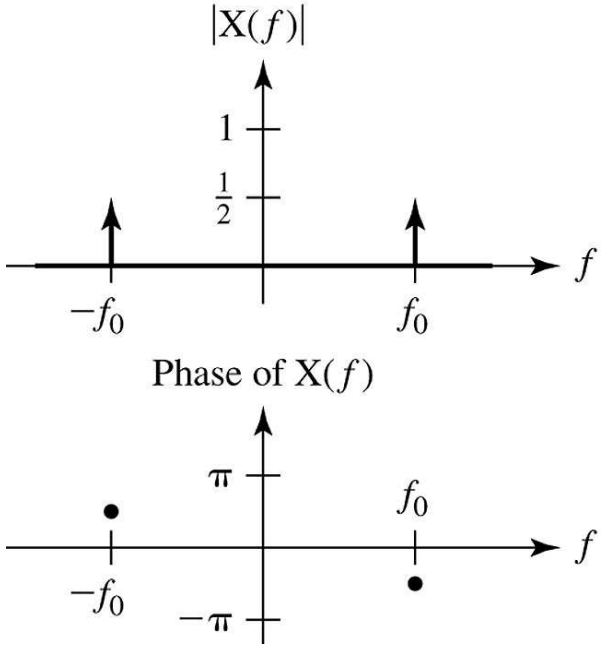
Fourier Transform of the sine function (1)

- The Fourier transform of $\sin(2\pi f_0 t)$ is:

$$x(t) = \sin(2\pi f_0 t) = \frac{1}{2j}e^{j2\pi f_0 t} - \frac{1}{2j}e^{-j2\pi f_0 t} = \frac{j}{2}e^{-j2\pi f_0 t} - \frac{j}{2}e^{j2\pi f_0 t}$$

$$\Rightarrow \quad X(f) = \frac{j}{2}\delta(f + f_0) - \frac{j}{2}\delta(f - f_0)$$

- $x(t)$ is real and odd and $X(f)$ is odd and pure imaginary.
- $|X(f)|$ is even and $\arg[X(f)]$ is odd.
- $x(0) = \sin(0) = 0$ and $\int_{-\infty}^{+\infty} X(f)\mathrm{d}f = \frac{j}{2} - \frac{j}{2} = 0$ implying that $x(0) = \int_{-\infty}^{+\infty} X(f)\mathrm{d}f$.



Fourier Transform of the sine function (2)

- Note that $X(f) = \frac{j}{2}\delta(f + f_0) - \frac{j}{2}\delta(f - f_0)$ implying that:

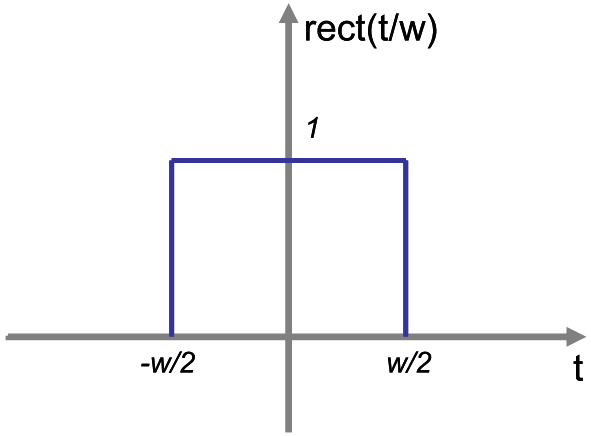
$$\begin{aligned} X(-f) &= \frac{j}{2}\delta(-f + f_0) - \frac{j}{2}\delta(-f - f_0) \\ &= \frac{j}{2}\delta\left(-(f - f_0)\right) - \frac{j}{2}\delta\left(-(f + f_0)\right) \\ &= \frac{j}{2}\delta(f - f_0) - \frac{j}{2}\delta(f + f_0) = -X(f) \end{aligned}$$

  implying that $X(f)$, the Fourier transform of $\sin(2\pi f_0 t)$, is an odd function of f.

  - Note that the following relation was used: $\delta(a(f - f_0)) = \frac{1}{|a|}\delta(f - f_0)$ where $a = -1$.

Example 1 (1)

- Determine the Fourier transform of the function $\mathrm{rect}\left(\frac{t}{w}\right)$



$$X(f) = \int_{-\infty}^{+\infty} x(t)e^{-j2\pi ft}\mathrm{d}t = \int_{-\frac{w}{2}}^{+\frac{w}{2}} e^{-j2\pi ft}\mathrm{d}t = -\frac{1}{j2\pi f}\left[e^{-j2\pi ft}\right]_{-\frac{w}{2}}^{+\frac{w}{2}}$$

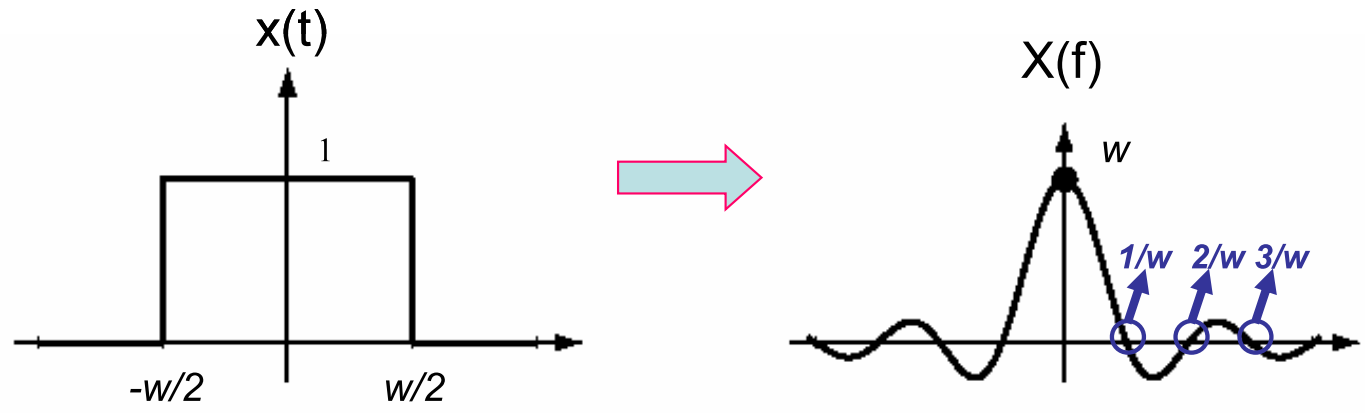
$$= \frac{1}{\pi f}\left[\frac{e^{j\pi fw} - e^{-j\pi fw}}{2j}\right] = \frac{1}{\pi f}\sin(\pi fw) = w\,\mathrm{sinc}(wf)$$

- Note that $X(f) = 0$ when f is a nonzero integer multiple of $1/w$.

Example 1 (2)

- Therefore:

$$\mathcal{F} : x(t) = \text{rect}\left(\frac{t}{w}\right) \quad \rightarrow \quad X(f) = w\,\text{sinc}(wf)$$

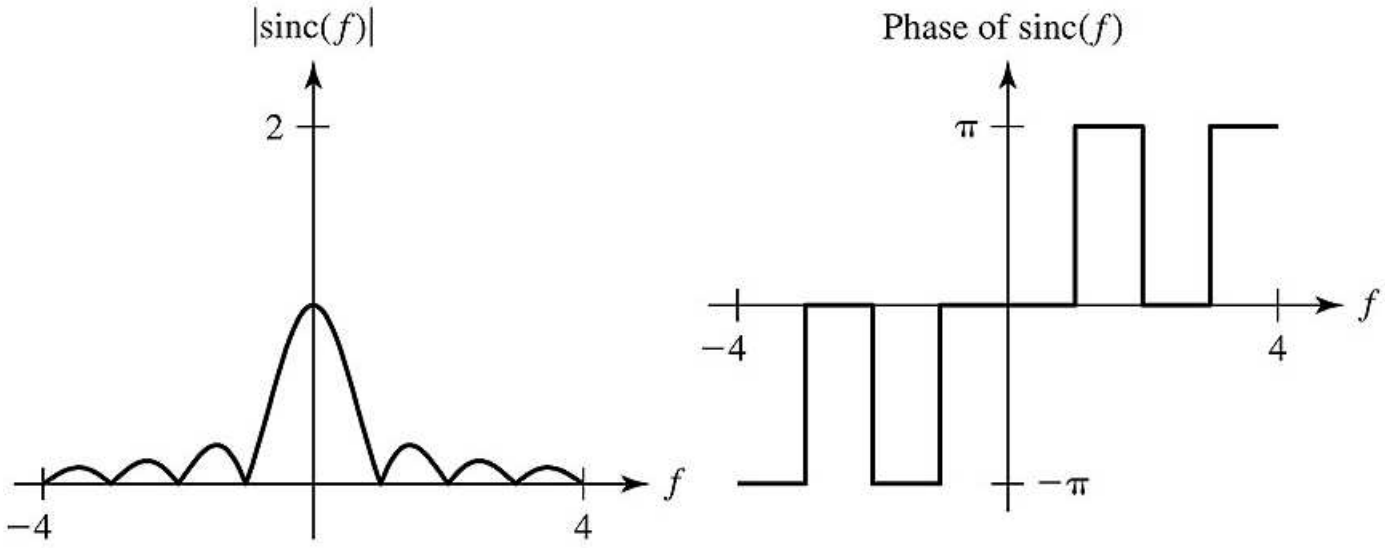


- Note the inverse relationship between the 2 widths (in the time domain and the frequency domain).
  - Signals having short durations occupy large bandwidths.
- In the limit, as $w \rightarrow +\infty$, $X(f)$ will tend to the unit impulse.

Example 1 (3)

- The next figure shows the magnitude and the phase spectra of the signal $x(t) = \mathrm{rect}(t)$:

$$X(f) = \mathcal{F}[x(t)] = \mathrm{sinc}(f)$$



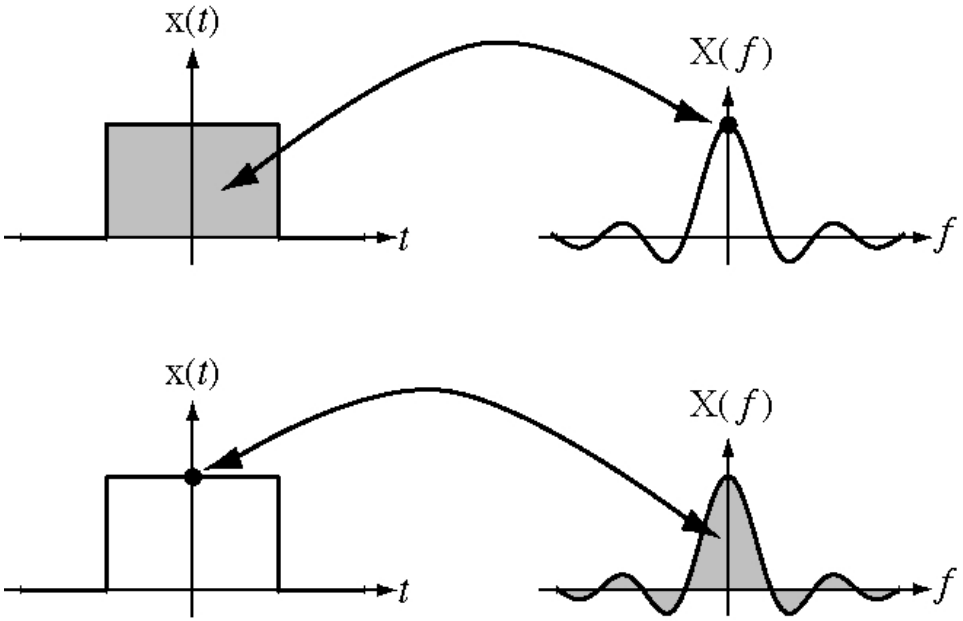
  - Note that $\mathrm{rect}(t)$ is real and even and, consequently, $\mathrm{sinc}(f)$ is also real and even.
  - Note that the magnitude spectrum is even while the phase spectrum is odd (since $x(t)$ is real).

Example 1 (4)

- Since $\mathcal{F}[\mathrm{rect}(t)] = \mathrm{sinc}(f)$, then applying the total area property of the FT implies that:

$$\int_{-\infty}^{+\infty} \mathrm{rect}(t)\mathrm{d}t = \mathrm{sinc}(0) = 1$$

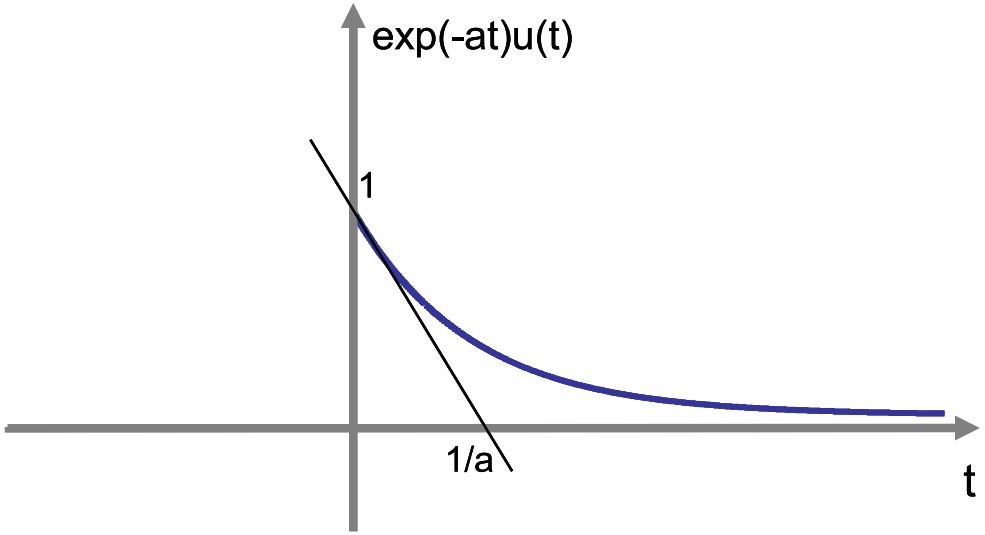
$$\int_{-\infty}^{+\infty} \mathrm{sinc}(f)\mathrm{d}f = \mathrm{rect}(0) = 1$$



Example 2 (1)

- Determine the Fourier transform of the exponential function:

$$x(t) = e^{-at}u(t) \quad , \quad a > 0$$



$$X(f) = \int_{-\infty}^{+\infty} x(t)e^{-j2\pi ft}\mathrm{d}t = \int_{0}^{+\infty} e^{-at}e^{-j2\pi ft}\mathrm{d}t = \int_{0}^{+\infty} e^{-(a+j2\pi f)t}\mathrm{d}t$$

$$= -\frac{1}{a+j2\pi f}\left[e^{-(a+j2\pi f)t}\right]_0^{+\infty} = -\frac{1}{a+j2\pi f}[0-1] = \frac{1}{a+j2\pi f}$$

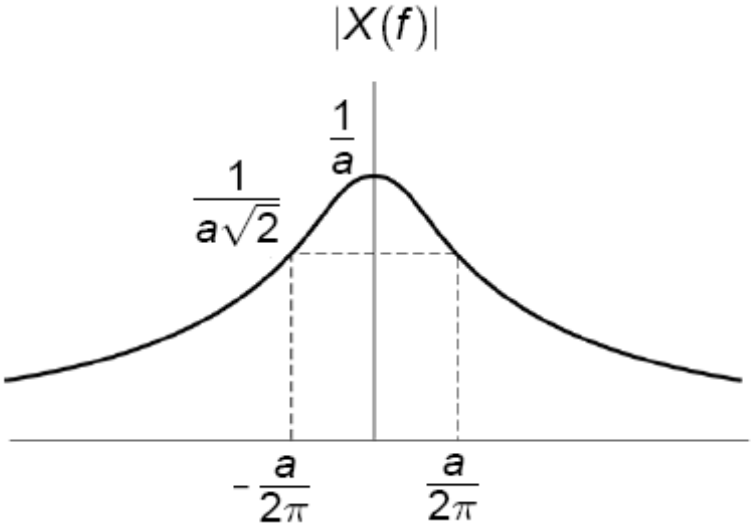
Example 2 (2)

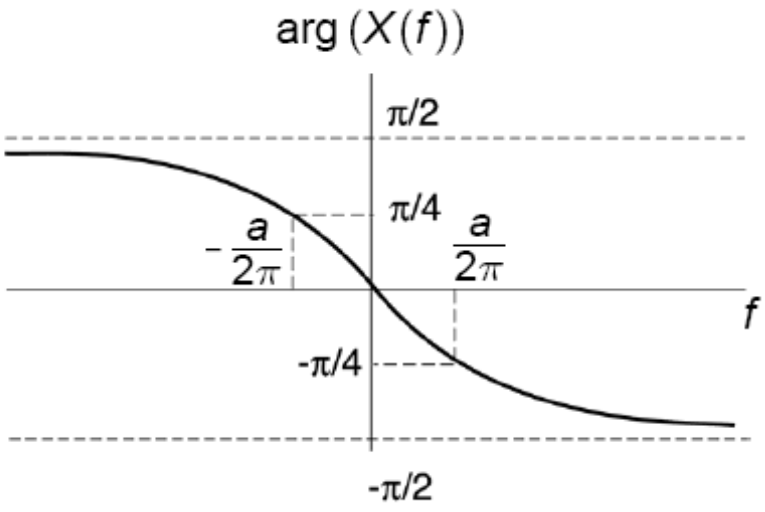
- Finally $X(f)$ is given by:

$$X(f) = \frac{1}{a + j2\pi f}$$

implying that:

$$|X(f)| = \frac{1}{\sqrt{a^2 + (2\pi f)^2}} \quad ; \quad \arg[X(f)] = -\tan^{-1}\left(\frac{2\pi f}{a}\right)$$





Example 2 (3)

- Note that when $x(t)$ decays slowly (a is small), $|X(f)|$ becomes more confined.
- Note also that:
  - Since the signal is real, then its magnitude spectrum is even and its phase spectrum is odd.
  - $x(t)$ is neither even nor odd implying that $X(f)$ is neither real nor pure imaginary.

Fourier Transform of periodic signals

- Consider a periodic signal with period T_0 (and fundamental frequency $f_0 = \frac{1}{T_0}$). The Fourier series representation of $x(t)$ is:

$$x(t) = \sum_{k=-\infty}^{+\infty} X[k] e^{j2\pi(kf_0)t}$$

- Given that:

$$e^{j2\pi(kf_0)t} \Leftrightarrow \delta(f - kf_0)$$

- Therefore:

$$x(t) = \sum_{k=-\infty}^{+\infty} X[k] e^{j2\pi(kf_0)t}$$

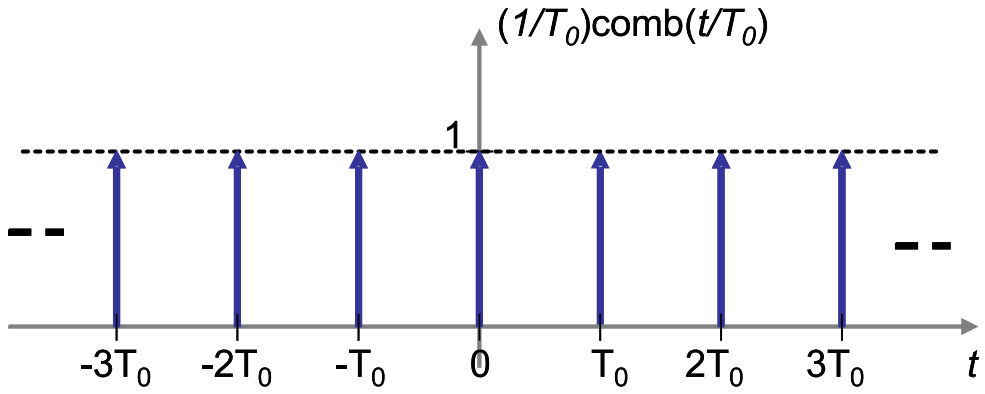
$$\Updownarrow$$

$$X(f) = \sum_{k=-\infty}^{+\infty} X[k] \delta(f - kf_0)$$

Example 3 (1)

- Calculate the Fourier Transform of the sampling function:

$$x(t) = \frac{1}{T_0}\text{comb}\left(\frac{t}{T_0}\right) = \sum_{n\in\mathbb{Z}} \delta(t - nT_0)$$



- *Step 1*: Determine the Fourier series coefficients of $x(t)$:

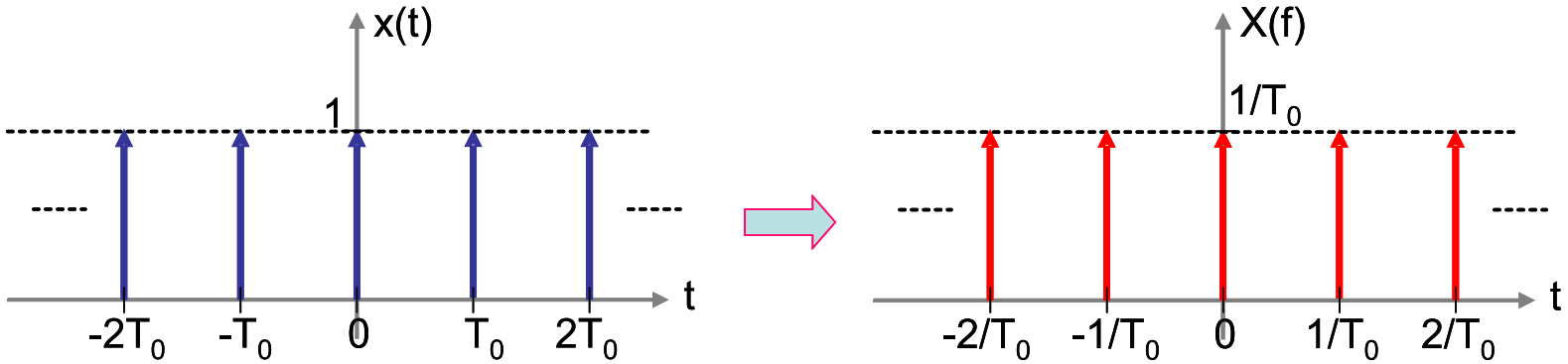
$$X[k] = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)e^{-j2\pi(kf_0)t}\mathrm{d}t \quad ; \quad f_0 = \frac{1}{T_0}$$

$$= \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \underbrace{\delta(t)e^{-j2\pi(kf_0)t}}_{=\delta(t)}\mathrm{d}t = \frac{1}{T_0} \quad \forall \ k$$

Example 3 (2)

- Consequently:
  $X(f) = \sum_{k\in\mathbb{Z}} X[k]\delta(f - kf_0) = \frac{1}{T_0}\sum_{k\in\mathbb{Z}} \delta\left(f - \frac{k}{T_0}\right)$.
- As a conclusion:

$$x(t) = \sum_{n\in\mathbb{Z}} \delta(t - nT_0) \Leftrightarrow X(f) = \frac{1}{T_0}\sum_{n\in\mathbb{Z}} \delta\left(f - \frac{n}{T_0}\right)$$



- Note that:
  - Period in time domain: T_0.
  - Period in frequency domain: $\frac{1}{T_0}$.
  - The inverse relationship again!

Example 3 (3)

- Note that $X(f)$ can be written as:

$$X(f) = \frac{1}{T_0}\sum_{n\in\mathbb{Z}}\delta\left(f - \frac{n}{T_0}\right) = \frac{1}{T_0}\left[T_0 \mathrm{comb}\left(T_0 f\right)\right] = \mathrm{comb}\left(T_0 f\right)$$

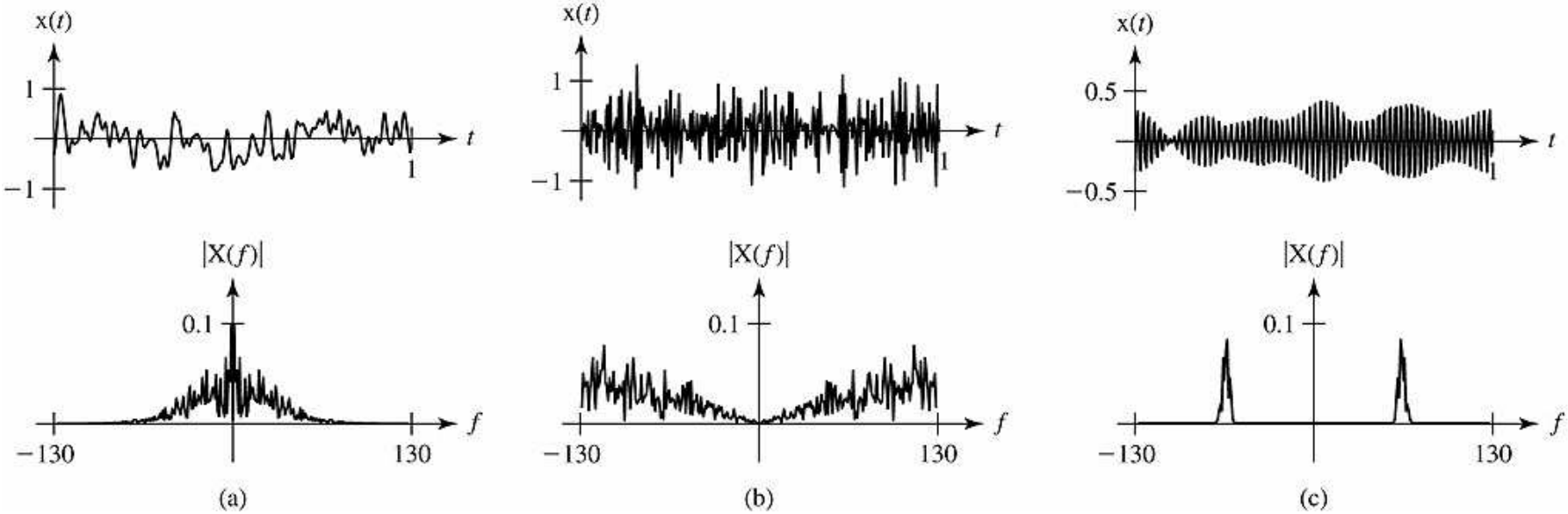
implying that:

$$\frac{1}{T_0}\mathrm{comb}\left(\frac{t}{T_0}\right) \quad \Leftrightarrow \quad \mathrm{comb}\left(T_0 f\right)$$

- In particular:

$$\mathcal{F}\left[\mathrm{comb}\left(t\right)\right] = \mathrm{comb}\left(f\right)$$

Examples of signals with different spectral contents



- It is important to develop an intuitive understanding of the relationship between the time variation of a signal and the frequency variation of its FT.
  - Figure (a) is an example of lowpass signals: these are low-frequency signals that vary slowly. They are characterized by their smoothness.
  - Figure (b) is an example of highpass signals: these are characterized by their rapid variations.
  - Figure (c) is an example of bandpass signals: these are characterized by their similarities to sinusoids.

Summary

- As a summary of some common Fourier Transforms:

$$
\begin{aligned}
\delta(t) &\rightarrow 1 \\
1 &\rightarrow \delta(f) \\
e^{j2\pi f_0 t} &\rightarrow \delta(f - f_0) \\
\cos(2\pi f_0 t) &\rightarrow \frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0) \\
\sin(2\pi f_0 t) &\rightarrow \frac{j}{2}\delta(f + f_0) - \frac{j}{2}\delta(f - f_0) \\
\mathrm{rect}(t) &\rightarrow \mathrm{sinc}(f) \\
\mathrm{comb}(t) &\rightarrow \mathrm{comb}(f) \\
e^{-at}u(t) &\rightarrow \frac{1}{a + j2\pi f} \quad ; \quad a > 0
\end{aligned}
$$